

Homework 02 — Model Selection and Prediction Error

Eric Chen · All computations in R (base); seeds 43202 (Q1) and 43203 (Q2).

Question 1 · Training and test error under a fixed design

$n = 100$, $p = 20$, $\beta_j = 0.4^{\sqrt{j}}$, $\mu = \mathbf{X}_{\text{all}}\beta$. One realized $\mathbf{X}_{\text{all}}$ is drawn after `set.seed(43202)` and then held fixed; 200 independent pairs $(\mathbf{y}, \mathbf{y}^*)$ are generated from it.

```
set.seed(43202)
X_all <- matrix(rnorm(n * p), n, p); beta <- 0.4^sqrt(1:p); mu <- as.vector(X_all %*% beta)
for (s in 1:200) { y <- mu + rnorm(n); ystar <- mu + rnorm(n)
  for (m in 0:p) { Xm <- if (m > 0) cbind(1, X_all[, 1:m, drop = FALSE]) else matrix(1, n, 1)
    yhat <- as.vector(Xm %*% qr.solve(Xm, y))
    train_mat[s, m+1] <- mean((y - yhat)^2); test_mat[s, m+1] <- mean((ystar - yhat)^2) } }
```

(a) Average training and test MSE

The two averaged curves are the solid lines in Figure 1. Average training MSE falls monotonically from 1.315 at $m = 0$ to 0.795 at $m = 20$, while average test MSE first falls to a minimum of **1.094 at $m = 6$** and then rises steadily back to 1.217 at $m = 20$ — the classic U shape.

Monotonicity check. For every one of the 200 runs I formed the 20 successive differences $\text{MSE}_{\text{train},m+1} - \text{MSE}_{\text{train},m}$ (4,000 differences in total). The largest value observed was -4.13×10^{-11} , i.e. **every** difference was ≤ 0 and no run showed an increase. This is not luck: the column space of $\mathbf{X}_m$ is nested inside that of $\mathbf{X}_{m+1}$, and least squares minimizes $\|\mathbf{y} - \mathbf{X}_m \mathbf{b}\|_2^2$ over that space, so enlarging the space can only lower the attained minimum.

(b) Theoretical expectations

With $\mathbf{H}_m$ the hat matrix and $B_m^2 = \|\mathbf{I}_n - \mathbf{H}_m\|_2^2$ and $\sigma^2 = 1$, the two conditional expectations were evaluated exactly and added as dashed lines in Figure 1.

m	B_m^2/n	avg train	$E(\text{train} \mid \mathbf{X})$	avg test	$E(\text{test} \mid \mathbf{X})$	one run, test
0	0.3074	1.3147	1.2974	1.3367	1.3174	1.0024
1	0.1963	1.1881	1.1763	1.2279	1.2163	0.9051
2	0.1172	1.0986	1.0872	1.1546	1.1472	0.8537
3	0.0719	1.0415	1.0319	1.1144	1.1119	0.7654
4	0.0522	1.0105	1.0022	1.1046	1.1022	0.8864
5	0.0350	0.9806	0.9750	1.0979	1.0950	0.8559
6	0.0247	0.9623	0.9547	1.0940	1.0947	0.8090
8	0.0143	0.9327	0.9243	1.1034	1.1043	0.8195
10	0.0092	0.9061	0.8992	1.1186	1.1192	0.8328
15	0.0019	0.8524	0.8419	1.1594	1.1619	0.9206
20	0.0000	0.7948	0.7900	1.2170	1.2100	0.9590

Comparison. The simulation averages track the theory closely: the largest absolute discrepancy over all 21 values of m is 0.017 for training MSE and 0.019 for test MSE, both consistent with Monte Carlo error from only 200 runs (the standard error of an averaged MSE is roughly $\sqrt{(2/n)/\sqrt{200}} \approx 0.010$). The theoretical test curve is minimized at the same $m = 6$. The decomposition is visible in the table: B_m^2/n falls from 0.307 to exactly 0 (at $m = 20$ the model contains μ in its column space), while the estimation term $(m+1)/n$ grows linearly from 0.01 to 0.21. Training MSE subtracts that term and test MSE adds it, so the two dashed curves are symmetric about the common bias curve $B_m^2/n + 1$ and their vertical gap is exactly the optimism $2(m+1)/n$.

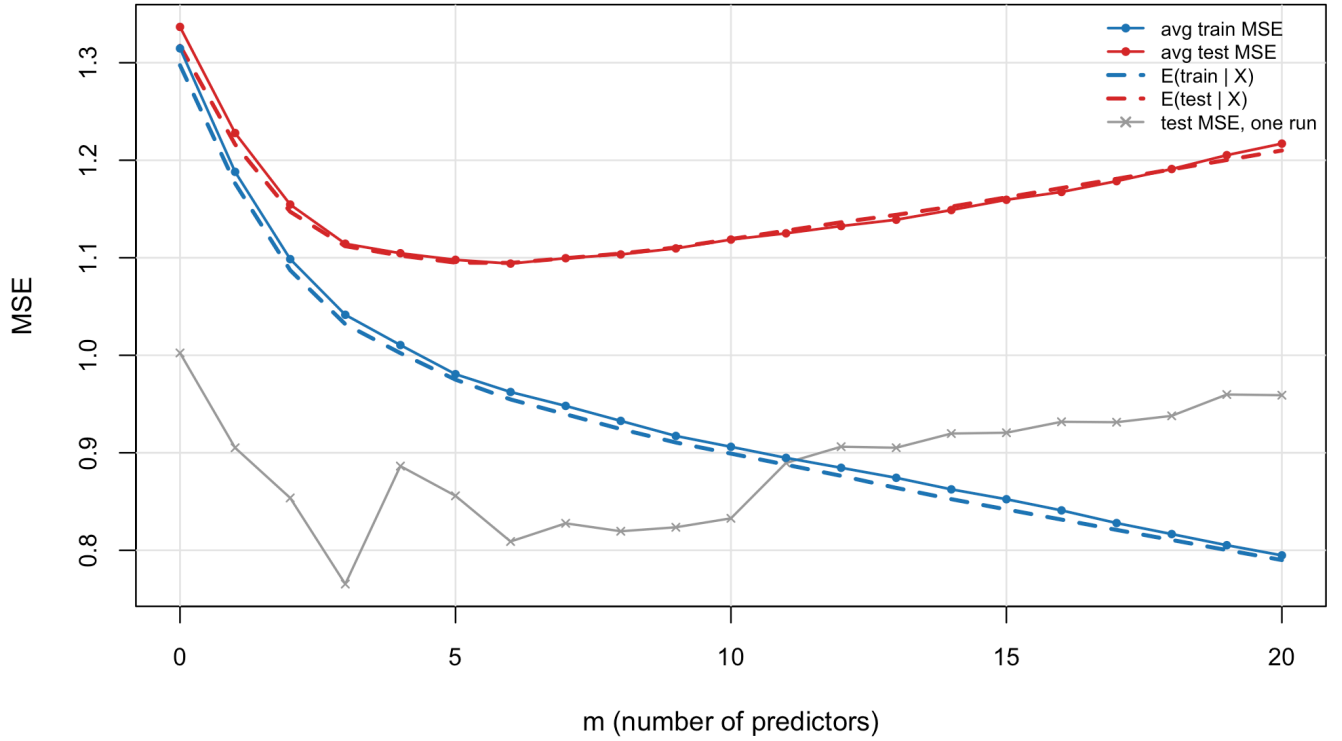


Figure 1. Average training and test MSE over 200 runs (solid, filled circles), the conditional expectations of part (b) (dashed), and the test MSE from a single run (grey, crosses). Average test MSE is minimized at $m = 6$; the single-run curve happens to be minimized at $m = 3$.

(c) One run, and why training MSE cannot choose m

Why the single curve is rough. The grey curve is one draw of $\text{MSE}_{\text{test},m}$, not an expectation. Writing $\mathbf{y}^* - \mathbf{H}_m \mathbf{y} = (\mathbf{I} - \mathbf{H}_m) \boldsymbol{\mu} + \boldsymbol{\varepsilon}^* - \mathbf{H}_m \boldsymbol{\varepsilon}$, a single realization equals its expectation plus fluctuation terms in $\boldsymbol{\varepsilon}$ and $\boldsymbol{\varepsilon}^*$ whose standard deviation is of order $\sqrt{(2/n)} \approx 0.14$. That is *larger* than the entire range of the expected test curve (1.094 to 1.337, a span of 0.24), so the noise is not a small perturbation of the signal — it dominates locally. Averaging 200 independent runs shrinks this noise by a factor $\sqrt{200} \approx 14$, to about 0.010, which is why the red curve is smooth and the grey one is not. Concretely, the standard deviation of successive increments is 0.047 for the single run versus 0.032 for the average, and the increments of the single run change sign repeatedly (e.g. the dip to 0.765 at $m = 3$ followed by a jump to 0.886 at $m = 4$). Note also that the whole grey curve sits below the red one: this run simply drew a small $\boldsymbol{\varepsilon}^*$, and the same vertical offset shifts every point, since all 21 models are evaluated on one shared pair $(\mathbf{y}, \mathbf{y}^*)$. That common shift is exactly why a single test curve can be badly misleading about the *level* of prediction error, while its argmin (here 3, not 6) is misleading about the *choice*.

Why training MSE cannot select the predictor count. Compare the two expectations,

$$E(\text{MSE}_{\text{train},m} | \mathbf{X}) = \frac{B_m^2}{n} + \sigma^2 - \sigma^2 \frac{m+1}{n}, \quad E(\text{MSE}_{\text{test},m} | \mathbf{X}) = \frac{B_m^2}{n} + \sigma^2 + \sigma^2 \frac{m+1}{n}.$$

Both contain the same approximation-bias term B_m^2/n , which is non-increasing in m because the column spaces are nested. They differ only in the sign of the estimation-variance term $\sigma^2(m+1)/n$. In the training expectation that term enters with a *minus* sign, so it is a reward rather than a penalty: **both** terms of the training expectation are non-increasing in m , and expected training MSE therefore decreases monotonically and is always minimized by the largest model (0.790 at $m = 20$ in the table). Training MSE measures how well the fit reproduces the very data used to compute the coefficients, so the $m+1$ fitted parameters absorb part of the noise $\boldsymbol{\varepsilon}$ itself — that is why the residual variance shrinks by the factor $1 - (m+1)/n$. In the test expectation the same quantity enters with a *plus* sign, because on fresh data the estimation noise $\mathbf{H}_m \boldsymbol{\varepsilon}$ is independent of $\boldsymbol{\varepsilon}^*$ and can only add error. The trade-off between falling bias and rising variance is what creates the interior minimum at $m = 6$, and it is invisible to the training criterion. The gap between the curves, the optimism

$2\sigma^2(m+1)/n$, itself grows with m (0.02 at $m = 0$ up to 0.42 at $m = 20$), so training MSE is not merely biased downward — it is biased downward by an amount that increases with model size, which is precisely the direction that corrupts a comparison. Any usable criterion must add back a penalty of that order, which is exactly what C_p , AIC and BIC do in Questions 3 and 4.

Question 2 · Prediction error at one target point

$n = 100$, $p = 6$, $\sigma^2 = 1$, $\beta = (0.5, \dots, 0.5)^\top$, target $\mathbf{x}_0 = (0, 1, 0, 0, 1, 0)^\top$. The cosine design satisfies $\mathbf{1}^\top \mathbf{X}_{\text{all}} = \mathbf{0}^\top$ and $n^{-1} \mathbf{X}_{\text{all}}^\top \mathbf{X}_{\text{all}} = \mathbf{I}_p$.

(a) Theoretical bias, variance and squared error

$\mu_0 = \mathbf{x}_0^\top \beta = 0.5 + 0.5 = 1$. Evaluating the given formula for $m = 0, \dots, 6$:

m	0	1	2	3	4	5	6
squared bias $(\sum_{j>m} x_{0j} \beta_j)^2$	1.00	1.00	0.25	0.25	0.25	0.00	0.00
variance $(\sigma^2/n)(1 + \sum_{j \leq m} x_{0j}^2)$	0.01	0.01	0.02	0.02	0.02	0.03	0.03
expected squared error	1.01	1.01	0.27	0.27	0.27	0.03	0.03
simulation average (200 runs)	1.0189	1.0189	0.2671	0.2671	0.2671	0.0338	0.0338

Why the formula has these two pieces. The prediction is $\hat{\mu}_{0,m} = \beta_0 + \sum_{j \leq m} x_{0j} \hat{\beta}_j$, and its error splits into a systematic part and a random part. The **squared-bias term** is what the model cannot represent: the omitted predictors $j > m$ still contribute $x_{0j} \beta_j$ to the true mean μ_0 , but the fitted model has no coefficient for them, so their entire contribution is missing from every fit no matter how much data is collected. Because the design is orthogonal, dropping a column does not distort the surviving coefficients — each $\hat{\beta}_j$ stays unbiased for β_j — so the bias is exactly the sum of the omitted contributions. The **variance term** is the cost of estimating the $m+1$ coefficients the model does keep: with $n^{-1} \mathbf{X}^\top \mathbf{X} = \mathbf{I}$ each $\hat{\beta}_j$ (and the intercept) has variance σ^2/n and they are mutually uncorrelated, so the variance of the linear combination is $(\sigma^2/n)(1 + \sum_{j \leq m} x_{0j}^2)$, the “1” being the intercept. Adding a predictor thus trades a fixed reduction in bias for a fixed increase in variance.

Where the error changes and where it does not. The expected error changes only at $m = 2$ and $m = 5$ ($1.01 \rightarrow 0.27$ and $0.27 \rightarrow 0.03$) and is unchanged at $m = 1, 3, 4$ and 6 . The reason is that $x_{0j} = 0$ for $j \in \{1, 3, 4, 6\}$: adding such a predictor removes nothing from the bias sum (its term $x_{0j} \beta_j = 0$ was already absent) and adds nothing to the variance sum (its contribution $x_{0j}^2 = 0$). Both effects are exactly zero, so the criterion is flat there.

(b) Simulation with seed 43203

```
set.seed(43203)
for (s in 1:200) { y <- as.vector(X_all %*% beta + rnorm(n))
  for (m in 0:p) { Xm <- if (m > 0) cbind(1, X_all[, 1:m, drop = FALSE]) else matrix(1, n, 1)
    bh <- qr.solve(Xm, y); x0m <- c(1, if (m > 0) x0[1:m])
    sq[s, m+1] <- (sum(x0m * bh) - mu0)^2 } }
```

The simulation averages are in the last row of the table above and are plotted against the theory in Figure 2. They agree to within Monte Carlo error at every m (largest discrepancy 0.009 at $m = 0$, where the squared error itself is a noisy squared quantity centred at 1.01), and they reproduce the flat-then-drop staircase exactly. **The smallest m minimizing the theoretical error is $m = 5$** , with expected squared error 0.03; $m = 6$ attains the same value but uses one more parameter, so the parsimonious choice is 5.

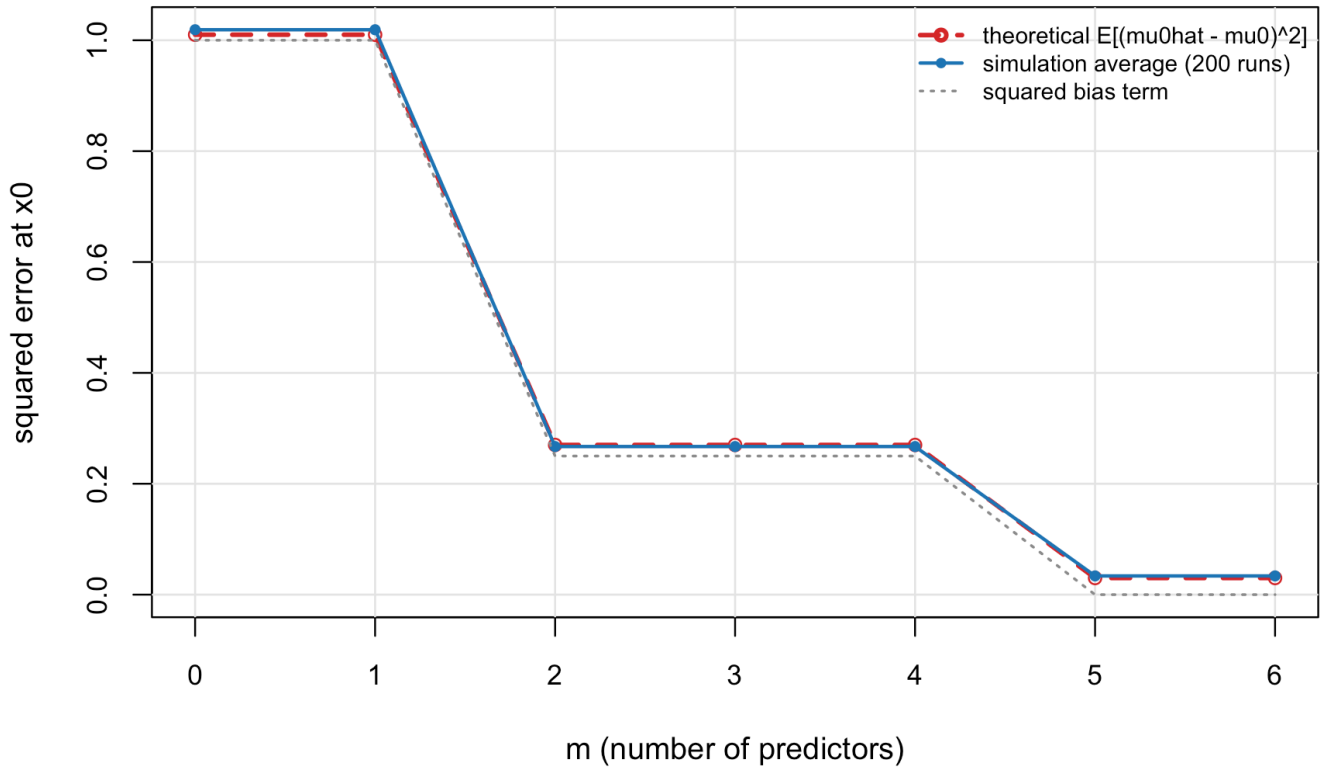


Figure 2. Theoretical expected squared error at $\mathbf{x}_0$ (dashed), the average of 200 simulated squared errors (solid), and the squared-bias component alone (dotted). The curves are flat except at $m = 2$ and $m = 5$, the two indices where $x_{0j} \neq 0$.

(c) Why only predictors 2 and 5 matter here

All six coefficients are nonzero, but the target is a *single linear functional* of them, $\mu_0 = \mathbf{x}_0^\top \boldsymbol{\beta} = \sum_j x_{0j} \beta_j$, and each β_j enters weighted by x_{0j} . Since $\mathbf{x}_0$ has zeros in coordinates 1, 3, 4 and 6, those four coefficients are multiplied by zero: they are genuinely part of the data-generating mechanism and they matter a great deal at *other* points of the design, but they contribute nothing to the mean response *at this particular* $\mathbf{x}_0$. Orthogonality is what makes the argument clean — it guarantees that omitting column j neither biases the retained β_k nor changes their variances, so there is no indirect route by which predictor 1, 3, 4 or 6 could influence the error at $\mathbf{x}_0$. In a correlated design an omitted predictor would generally shift the retained coefficients and would matter even with $x_{0j} = 0$.

Contrast with Question 1. Question 1’s test MSE averages the squared error over all n rows of the design, so it is a weighted average of target-specific errors over the whole covariate distribution. In that aggregate criterion no direction is ignored: every nonzero β_j reduces bias somewhere in the sample, and every extra parameter costs σ^2/n uniformly, which is why the Question 1 curve moves at *every* m and produces a smooth interior minimum. Here the cost–benefit calculation is local: it is driven only by the coordinates in which $\mathbf{x}_0$ actually loads, so the curve is a step function and four of the six predictors are free to include and free to omit. The practical lesson is that the “best” model size is defined relative to the loss one intends to minimize — a model can be optimal for predicting at one point and clearly suboptimal for average prediction over the design, and conversely.

Question 3 · The optimism correction

(a) Derivation of the two expectations

Throughout, $\mathbf{H}$ is symmetric and idempotent, so $\mathbf{I} - \mathbf{H}$ is too, $(\mathbf{I} - \mathbf{H})\mathbf{H} = \mathbf{0}$, $\text{tr}(\mathbf{H}) = p+1$ and $\text{tr}(\mathbf{I} - \mathbf{H}) = n - (p+1)$.

Training error. Substituting $\mathbf{y} = \boldsymbol{\mu} + \boldsymbol{\varepsilon}$,

$$\mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})(\boldsymbol{\mu} + \boldsymbol{\varepsilon}) = (\mathbf{I} - \mathbf{H})\boldsymbol{\mu} + (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon},$$

so, expanding the squared norm and using idempotence of $\mathbf{I} - \mathbf{H}$,

$$\|\mathbf{y} - \mathbf{H}\mathbf{y}\|_2^2 = B^2 + 2\boldsymbol{\mu}^\top (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon} + \boldsymbol{\varepsilon}^\top (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}.$$

Conditionally on $\mathbf{X}$ the middle term has mean zero because $E(\boldsymbol{\varepsilon} | \mathbf{X}) = \mathbf{0}$, and the quadratic form has mean $\sigma^2 \text{tr}(\mathbf{I} - \mathbf{H})$. Hence

$$E(\|\mathbf{y} - \mathbf{H}\mathbf{y}\|_2^2 | \mathbf{X}) = B^2 + \sigma^2 \{n - (p+1)\} \Rightarrow E(\text{MSE}_{\text{train}} | \mathbf{X}) = \frac{B^2}{n} + \sigma^2 \left(1 - \frac{p+1}{n}\right).$$

Test error. Now $\mathbf{y}^* = \boldsymbol{\mu} + \boldsymbol{\varepsilon}^*$ is independent of $\mathbf{y}$, and $\mathbf{H}\mathbf{y} = \mathbf{H}\boldsymbol{\mu} + \mathbf{H}\boldsymbol{\varepsilon}$, so

$$\mathbf{y}^* - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\boldsymbol{\mu} + \boldsymbol{\varepsilon}^* - \mathbf{H}\boldsymbol{\varepsilon}.$$

Expanding the squared norm gives the three squared terms plus three cross terms. Every cross term vanishes in expectation: $2\boldsymbol{\mu}^\top (\mathbf{I} - \mathbf{H})E(\boldsymbol{\varepsilon}^*) = 0$ and $-2\boldsymbol{\mu}^\top (\mathbf{I} - \mathbf{H})\mathbf{H}E(\boldsymbol{\varepsilon}) = 0$ because the errors have mean zero (the second also because $(\mathbf{I} - \mathbf{H})\mathbf{H} = \mathbf{0}$), and $-2E(\boldsymbol{\varepsilon}^{*\top} \mathbf{H}\boldsymbol{\varepsilon} | \mathbf{X}) = -2E(\boldsymbol{\varepsilon}^* | \mathbf{X})^\top \mathbf{H}E(\boldsymbol{\varepsilon} | \mathbf{X}) = 0$ by independence. For the squared terms, $E(\|\boldsymbol{\varepsilon}^*\|_2^2) = n\sigma^2$ and, using $\mathbf{H}^\top \mathbf{H} = \mathbf{H}$, $E(\|\mathbf{H}\boldsymbol{\varepsilon}\|_2^2 | \mathbf{X}) = E(\boldsymbol{\varepsilon}^\top \mathbf{H}\boldsymbol{\varepsilon} | \mathbf{X}) = \sigma^2 \text{tr}(\mathbf{H}) = \sigma^2(p+1)$. Therefore

$$E(\|\mathbf{y}^* - \mathbf{H}\mathbf{y}\|_2^2 | \mathbf{X}) = B^2 + n\sigma^2 + \sigma^2(p+1) \Rightarrow E(\text{MSE}_{\text{test}} | \mathbf{X}) = \frac{B^2}{n} + \sigma^2 \left(1 + \frac{p+1}{n}\right).$$

The structural difference is that in the training error the fitted noise $\mathbf{H}\boldsymbol{\varepsilon}$ is *subtracted from the same* $\boldsymbol{\varepsilon}$ that generated it (removing $\sigma^2(p+1)$ of variance), whereas in the test error it is measured against an independent $\boldsymbol{\varepsilon}^*$ and therefore *adds* $\sigma^2(p+1)$.

(b) Expected optimism

$$E(\text{MSE}_{\text{test}} - \text{MSE}_{\text{train}} | \mathbf{X}) = 2\sigma^2 \frac{p+1}{n}.$$

The bias term B^2/n cancels, so the optimism depends only on the number of fitted coefficients, the sample size and the noise level — not on how well the model approximates $\boldsymbol{\mu}$. With $n = 120$, $p = 4$, $B^2/n = 0.04$, $\sigma^2 = 1$, so that $(p+1)/n = 5/120 = 0.041667$:

quantity	expression	value
$E(\text{MSE}_{\text{train}} \mathbf{X})$	$0.04 + 1 - 5/120$	0.983333
$E(\text{MSE}_{\text{test}} \mathbf{X})$	$0.04 + 1 + 5/120$	1.066667
expected optimism	$2(1)(5)/120$	0.083333

Why the ordering can fail for a realized pair. The inequality $\text{MSE}_{\text{train}} < \text{MSE}_{\text{test}}$ is a statement about *expectations*, not about individual draws. For one realized pair $(\boldsymbol{\varepsilon}, \boldsymbol{\varepsilon}^*)$ the difference is a random variable whose leading fluctuation is $n^{-1}(\|\boldsymbol{\varepsilon}^*\|_2^2 - \|\boldsymbol{\varepsilon}\|_2^2)$, with standard deviation of order $2\sigma^2/\sqrt{n} = 2/\sqrt{120} \approx 0.18$. That is more than twice the mean gap of 0.083, so a test draw that happens to be quiet, or a training draw that happens to be noisy, easily reverses the sign; sign reversals in individual runs are common, not exotic. The guarantee only has force after averaging — which is exactly what Figure 1 shows, where the averaged curves separate cleanly but the single grey run does not respect the pattern at all.

(c) The corrected test MSE and Mallows' C_p

With $n = 120$, $p = 4$, $\text{RSS} = 116.4$ and $\sigma^2 = 0.96$:

$$\text{MSE}_{\text{test}}^{\hat{E}} = \frac{116.4 + 2(5)(0.96)}{120} = \frac{116.4 + 9.6}{120} = \frac{126.0}{120} = \mathbf{1.05}, \quad C_p = \frac{116.4}{0.96} - 120 + 2(5) = 121.25 - 120 + 10 = \mathbf{11.25}.$$

The relationship. Divide the first expression by σ^2 and rearrange:

$$MSE_{\text{test}} = \frac{\sigma^2}{n} (C_p + n) = \frac{\sigma^2}{n} C_p + \sigma^2, \quad \text{equivalently} \quad C_p = n \frac{MSE_{\text{test}} - \sigma^2}{\sigma^2}.$$

The numbers check out: $(0.96/120)(11.25) + 0.96 = 0.09 + 0.96 = 1.05$. So the two quantities are related by an **increasing affine map**, with slope $\sigma^2/n = 0.008 > 0$ and intercept σ^2 . Both are just RSS/n — the optimistic training error — plus the estimated optimism $2(p+1)\sigma^2/n$ derived in part (b), written on two different scales. When n and σ^2 are held common across all candidate models (which is why σ^2 is estimated once from a single large reference model rather than separately for each candidate), the slope and intercept are the same constants for every candidate, so an increasing affine map preserves order: whichever model has the smallest C_p also has the smallest MSE_{test} , and the two criteria produce identical rankings, not merely the same winner. Note the caveat implied by the phrase “common to all candidate models”: if each candidate used its own σ^2 , the map would differ between candidates and the equivalence would break.

Question 4 · Comparing Mallows’ C_p , AIC and BIC

Training data are rows 1–370 of `data/diabetes.csv` ($n = 370$); rows 371–442 are not used here.

```
tr <- read.csv("data/diabetes.csv")[1:370, ]
fA <- lm(y ~ bmi + bp + s5 + sex + s1 + s2 + s4, data = tr)
fB <- update(fA, . ~ . + s6); fF <- lm(y ~ ., data = tr)
s2h <- sum(residuals(fF)^2) / df.residual(fF)
```

(a) The three fits

model	predictors	p	RSS	residual df
Model A	bmi, bp, s5, sex, s1, s2, s4	7	1,098,908.6	362
Model B	A + s6	8	1,089,717.1	361
Full reference	all ten	10	1,089,451.7	359

The residual degrees of freedom of the full reference model are $n - (p+1) = 370 - 11 = \mathbf{359}$, giving the common noise-variance estimate

$$\sigma^2 = \frac{RSS_{\text{full}}}{n - (p+1)} = \frac{1,089,451.7}{359} = \mathbf{3034.684} \quad (\hat{\sigma} \approx 55.1 \text{ on the scale of } y).$$

(b) The three criteria

model	p	RSS	C_p	AIC ^{red}	BIC ^{red}
Model A	7	1,098,908.6	8.116	2974.640	3005.948
Model B	8	1,089,717.1	7.087	2973.532	3008.754
selects			B	B	A

C_p selects **Model B** ($7.087 < 8.116$), reduced AIC selects **Model B** ($2973.532 < 2974.640$), and reduced BIC selects **Model A** ($3005.948 < 3008.754$). As instructed, values are compared only within a column: the AIC and BIC columns differ by additive constants and their levels carry no meaning across criteria.

(c) Fit improvement versus the one-parameter penalty

Adding `s6` lowers RSS by $1,098,908.6 - 1,089,717.1 = 9,191.5$, a relative reduction of only 0.84%. Each criterion converts that gain to its own scale and compares it with the price of one extra parameter:

crit ^{erion}	gain from s6	gain	penalty for 1 parameter	penalty	net (B – A)
C_p	$\Delta\text{RSS} / \sigma^2$	3.029	2	2.000	–1.029 → B
AIC^{red}	$n \log(\text{RSS}_A/\text{RSS}_B)$	3.108	2	2.000	–1.108 → B
BIC^{red}	$n \log(\text{RSS}_A/\text{RSS}_B)$	3.108	$\log(370)$	5.914	+2.806 → A

Explaining the disagreement. The measured improvement is essentially the same number under all three criteria — about 3.0 to 3.1 units, and the two log-likelihood-based criteria share the identical gain of 3.108 because they use the identical goodness-of-fit term $n\log(\text{RSS}/n)$. The disagreement is therefore *entirely* a matter of the penalty. C_p and AIC both charge 2 units per parameter, which the gain of roughly 3.1 just exceeds, so both keep s6. BIC charges $\log(n) = \log(370) = 5.914$ units, nearly three times as much, and the gain does not come close to covering it, so BIC drops s6. The two verdicts are consistent with the ordinary t -statistic for s6 in Model B, $128.82/73.82 = 1.745$ ($P = 0.082$): the evidence is suggestive but well short of conventional significance. This is the expected behaviour of the criteria and not a defect — they answer different questions. AIC and C_p are estimates of prediction error and are willing to pay for a weak predictor if it is likely to help slightly out of sample; BIC approximates a posterior model probability, its penalty grows with n , and it is consistent for the smallest true model when one is in the candidate list. Since the margins here are around one AIC unit, the disagreement should be read as “the data do not distinguish A from B”, not as a contradiction.

Why selection does not establish truth. These criteria only rank the candidates they are given, on the relative scale of the assumed model class. Model B winning two of three criteria says it is the better of *two* specific linear models under an assumed homoscedastic normal-error linear structure with these ten measured predictors; if the true regression function is nonlinear, involves interactions, or depends on variables not in the data set, then every candidate is wrong and the criteria simply pick the least bad one. Moreover, selection is itself a random outcome: gaps of about 1 unit are well within the sampling variability of these statistics, and a different training sample of 370 patients could easily reverse the ranking. Finally, a criterion that is designed to minimize prediction error (AIC, C_p) makes no claim to recover the true model at all, and even BIC’s consistency guarantee holds only under the assumption that the true model is among the candidates.

Question 5 · Validation and final test data

(a) Where the procedure goes wrong

Steps 1 and 2 are legitimate. **Step 3 is the first misuse:** the moment the minimum over the eleven test MSEs determines *which* model is reported, the final test data have been used to make a modelling decision and are no longer held out. Step 4 then compounds this by reporting that same minimum as the error estimate, so the identical 72 observations are used both to select and to evaluate.

Each individual test MSE is an unbiased estimate of the prediction error of *its own* model, but the *minimum* of eleven such estimates is not unbiased for anything, and it is biased downward. The reason is a winner’s curse: each estimate equals a true error plus noise, and taking a minimum preferentially selects models whose noise happened to be most favourable on these particular 72 points. The selected model is thus doubly lucky — chosen because it looked good, and reported at the value that made it look good. The magnitude here is not academic. With only 72 test observations, the standard error of a single test MSE is about 399 (for $m = 9$, $\text{MSE} = 2369.3$), whereas the spread among the better candidates is only a few hundred:

m	0	1	2	3	4	5	6	7	8	9	10
test MSE (rows 371–442)	5823	5528	5527	3719	3247	3235	3208	2637	2625	2369	2485
5-fold CV MSE (rows 1–370)	5968	5801	5817	3976	3750	3773	3775	3227	3253	3139	3133

The winner ($m = 9$, 2369) beats its neighbour ($m = 10$, 2485) by 115, a gap far smaller than the noise in either number. Reporting 2369 as the expected future error of the whole select-then-report procedure is therefore optimistic: the procedure’s honest error is what a *freshly* chosen model achieves on *unseen* data, and repeating this experiment on a new test set would generally give a larger value and possibly a different winner.

(b) A correct analysis using five-fold cross-validation

1. **Split only the training data.** Partition the 370 training rows at random into five folds of about 74 rows each. The 72 final test rows are set aside and not touched.
2. **Cross-validate every candidate.** For each $m = 0, \dots, 10$ and each fold $k = 1, \dots, 5$, fit the m -predictor model on the other four folds and predict the held-out fold. Average the squared errors over all 370 held-out predictions to obtain $CV(m)$.
3. **Select the predictor count** as $\hat{m} = \operatorname{argmin}_m CV(m)$. This decision uses only rows 1–370. (A one-standard-error rule, choosing the smallest m within one CV standard error of the minimum, is a defensible alternative given how flat the curve is for $m \geq 7$.)
4. **Refit** the chosen model — the one with the first $\hat{m}$ predictors — on all 370 training rows, so that the final coefficients use the full training sample rather than 4/5 of it. The models fitted inside the folds are discarded; they existed only to produce $CV(m)$.
5. **Use the final test data exactly once, at the very end**, to evaluate that single refitted model. The resulting MSE is reported as an honest estimate of prediction error and is *not* allowed to change $\hat{m}$, the predictor order, or anything else. If it is used to revisit any choice, it stops being a test set.

Carried out on these data, CV is minimized at $\hat{m} = 10$ (CV = 3133, with $m = 9$ essentially tied at 3139); refitting on all 370 rows and evaluating once on rows 371–442 gives a final test MSE of **2485**. That is the number to report — larger than the 2369 the flawed procedure would have quoted, which is exactly the optimism described in (a).

(c) What can still be reported after the fact

Nothing needs to be hidden, but the labels must be honest. The analyst can still report, transparently and in full: the entire profile of eleven test MSEs, described as an *exploratory* look at how error varies with model size rather than as a validation; the fact that the minimum occurred at $m = 9$; and a clear statement that the test set has been used for selection, so that 2369 is a downward-biased estimate of future error and is not a valid final evaluation. The analyst can also still run the cross-validation of part (b) on rows 1–370 — those rows are uncontaminated by the peeking — and report $CV(\hat{m})$ as the best available honest estimate, together with the selected model and its coefficients. Reporting the number of models compared and the size of the gaps between them is part of that transparency.

What cannot be recovered is a clean final evaluation, because no amount of re-analysis can restore the independence that the peek destroyed. That requires **additional data never used for fitting or selection**: a fresh sample of patients from the same population, held out and evaluated once. If new data cannot be collected, the honest fallback is a nested resampling scheme applied to the pooled 442 observations — an outer split (or outer CV loop) that repeatedly holds out a block, performs the *entire* selection procedure including the choice of m inside the remaining data, and evaluates on the held-out block — which estimates the error of the procedure rather than of one pre-chosen model. This should be pre-specified before looking at the results, and it should be reported as a resampling estimate rather than as a true independent test.